

trum will face similar incentives depending on the way the rules are defined. The point is not that the most unrestricted environment is always the best. It's that our current system, without justification, assumes an exclusive licensing regime is the only viable answer.

## **Technologies Of Wireless Freedom**

There are three primary techniques for magnifying the efficiency of wireless devices in a shared environment: spread spectrum, cooperative networking and software-defined radio. These can all be used in licensed bands, though they reach their full potential in an unlicensed environment.

### **Spread Spectrum**

In a spread-spectrum system, wireless communications are digitized and chopped up into pieces, which are spread across a range of frequencies. If the receiver knows where to look, it can piece the message back together on the other end. Spread spectrum means that an individual frequency only carries a small part of each communication, so it's only occupied for a tiny slice of time. In the unlikely event that another message is occupying that slot, only that small portion of the signal must be re-sent. Spread spectrum was invented in the 1940s, and has been used extensively for military and other applications that require robustness and resistance to jamming or eavesdropping (because only the receiver knows how the signal is spread across the range of frequencies). Many mobile phone services use spread-spectrum today to improve efficiency within licensed bands, but the technique is even more powerful when used for underlay or in unlicensed bands.

### **Software-Defined Radio**

Every radio can be tuned to pick up a certain range of frequencies, and it takes some amount of time to change the tuning. Traditionally, these characteristics are fixed in the radio hardware. Thus, for example, the same radio can't pick up both FM radio and mobile phone transmissions, or both 2.4 Ghz and 5